ITI RESOURCE HUB - SUBJECT STUDY NOTES

Trade: ICTSM | Topic: TCP/IP Subnetting and VLSM Calculations

Class A default: /8 (255.0.0.0), Class B default: /16 (255.255.0.0), Class C default: /24 (255.255.255.0).

Borrowing Bits Formula: 2^n = Number of Subnets created (where n = borrowed network bits).

Host Capacity Formula: 2^h - 2 = Usable hosts per subnet (where h = remaining host bits).

Step 1: Determine the required host size for each departmental LAN segment.

Step 2: Allocate the largest subnet block first to avoid subnet overlapping.

Step 3: Document Network ID, Broadcast IP, First Usable Host IP, and Last Usable Host IP.